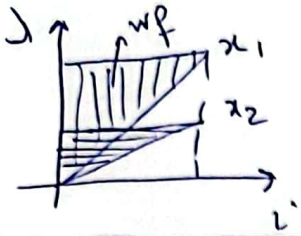


1) $B_g < B_c$ (0.75) $\rightarrow H_g \gg H_c$ (0.75) $\rightarrow H_g \gg H_c$ (0.75)



(ب) افزایش پتانسیل به معنی $\lambda \downarrow \rightarrow \lambda \downarrow \rightarrow \lambda \downarrow$ (0.5)
(ج) در وقت دوم سرعت به افزایش مقاومت میسر و کاهش تلفات (از این ناگاه) به افزایش مقاومت (1)

Total 4

2) (الف) $R_1 = \frac{7 + (7-3) \times 10^{-2}}{2000 \times 40 \times 10^{-7} \times 8 \times 10^{-4}} = 54709.5 \left(\frac{1}{H}\right)$

$R_2 = \frac{(6+6) \times 10^{-2}}{2000 \mu \times 4 \times 10^{-4}} = 119366.2 \left(\frac{1}{H}\right)$

$R_3 = \frac{3 \times 10^{-2}}{27000 \mu \times 8 \times 10^{-4}} = 11052.4 \left(\frac{1}{H}\right)$ (1) $\varphi = 0.4 \times 4 \times 10^{-4} = 16 \times 10^{-5} \text{ wb}$

$i = \frac{(R_1 + R_2 + R_3) \varphi}{300} = 0.099 \text{ A}$

ب) $R_{\text{tot}} \uparrow, \varphi = \frac{N_2 i}{R_{\text{tot}}} \downarrow \rightarrow B \downarrow$ (1)

Total 6

3) $R_1 = \frac{2 \times 10^{-3}}{\mu \times 16 \times 10^{-4} (1.02)} = 975.21 \times 10^3 \left(\frac{1}{H}\right)$

$R_2 = \frac{4 \times 10^{-3}}{\mu \times 16 \times 10^{-4} (1.02)} = 1950.43 \times 10^3 \left(\frac{1}{H}\right)$

$R_3 = \frac{3}{2} R_1 = 1462.82 \times 10^3 \left(\frac{1}{H}\right)$ (1) $R_{\text{eq}} = R_1 + R_2 \parallel R_3 = 1811.14 \times 10^3 \left(\frac{1}{H}\right)$

$\varphi_1 = \frac{N_1 i_1}{R_{\text{eq}}}, \varphi_3 = \frac{R_2}{R_2 + R_3} \varphi_1 = 0.571 \varphi_1 \rightarrow L_{13} = \frac{N_3 \varphi_3}{i_1} = \frac{N_1 N_3}{R_{\text{eq}}} \times 0.571 = 0.079 \text{ H}$

ب) $R_1 \uparrow \rightarrow R_{\text{eq}} \uparrow \rightarrow L_{13} \downarrow$ (0.5) (1) $wf = \int i d\lambda = \frac{0.04}{x+2} \int (\lambda^2 + 0.3 \lambda^{2.5}) d\lambda = \frac{0.04}{x+2} \left(\frac{\lambda^3}{3} + 0.3 \frac{\lambda^{3.5}}{3.5} \right) \Big|_0^3 = 0.237 \text{ J}$

$i = 0.249 \text{ A} \rightarrow wf = \lambda i - wf = 0.51 \text{ J}$ (0.5)

ب) $f_m = - \frac{\partial wf}{\partial x} \Big|_{\lambda=3} = \frac{0.04}{(x+2)^2} \left(\frac{\lambda^3}{3} + 0.3 \frac{\lambda^{3.5}}{3.5} \right) = 0.108 \text{ N}$ (1) $\rightarrow$ نیرو در راستای افزایش x (0.5)

Total 3.5